Great Salt Lake Land Cover Change

Comparing LCMS and NLCD land cover products from 1985-2024 over the Great Salt Lake region.

Generated March 13, 2026 at 01:47 PM

Executive Summary

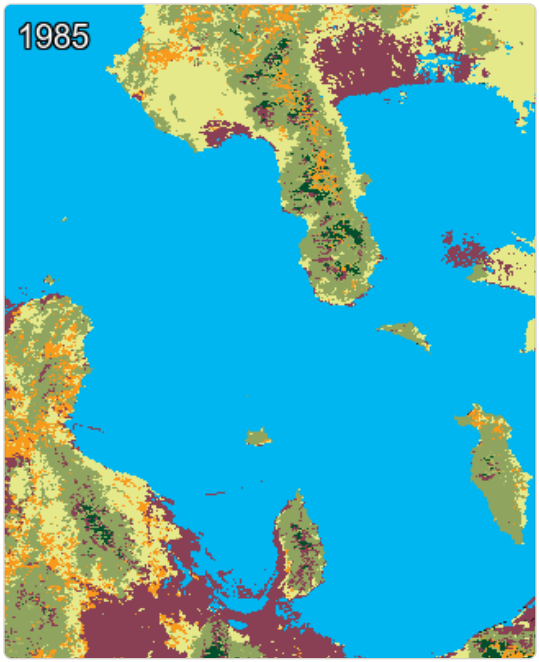
Executive Summary: Great Salt Lake Land Cover Change

This report analyzes land cover fluctuations in the Great Salt Lake region from 1985 to 2024 using data from the Landscape Change Monitoring System (LCMS) and the National Land Cover Database (NLCD). The primary trend identified across both datasets is a substantial reduction in water extent and a corresponding increase in barren surfaces. NLCD data characterizes the study area by a dynamic relationship between open water and barren land, with the former decreasing consistently over the 39-year period.

Transition analysis indicates that while the water class exhibited 53.76% stability between 1985 and 2000, the most frequent land cover conversion throughout the study period was the transition from water to barren or impervious surfaces. The contemporary landscape is dominated by three primary classes: open water, shrub/scrub, and barren land. LCMS data confirms that the loss of water area is the most frequent driver of land cover change in the region, as receding shorelines are replaced by exposed lakebed.

Spatial patterns visible in the 1984 and 1985 baseline maps show a large, contiguous body of water surrounded by a mosaic of shrub/scrub and barren land. The data indicates that land cover transitions are most concentrated along these perimeter interfaces. As the water extent contracted, the resulting land cover shifts occurred primarily at the lake's margins, where former aquatic areas converted to the barren land category.

LCMS Land Cover (1985-2024)

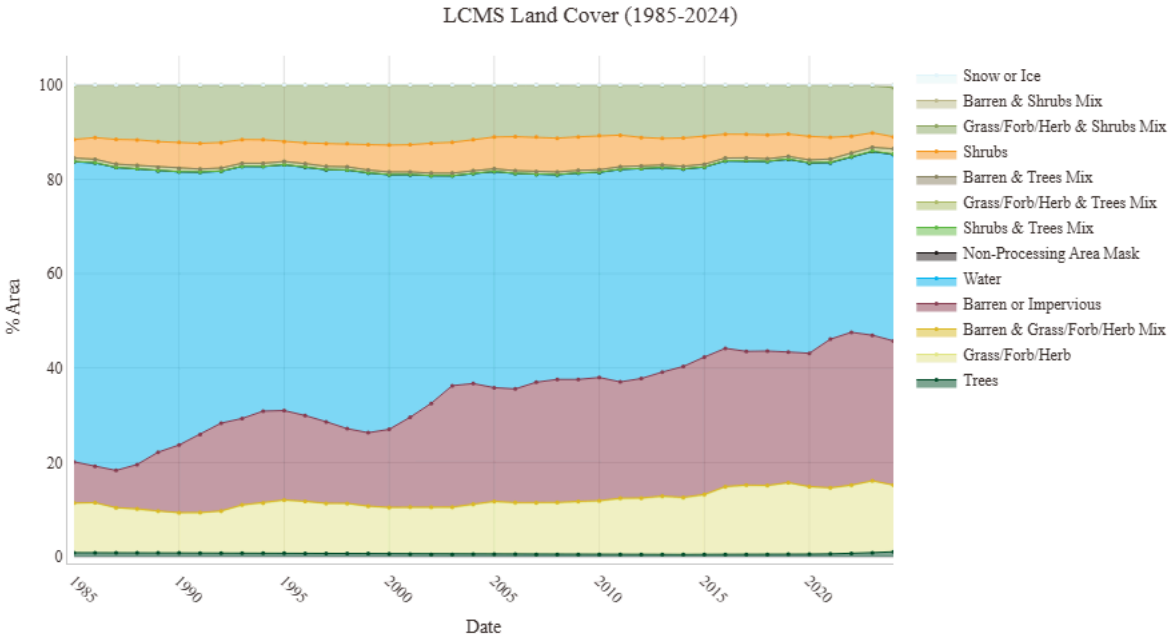


LCMS Land Cover (1985-2024)

The primary land cover trend between 1985 and 2024 is a substantial reduction in **Water** extent and a corresponding increase in **Barren or Impervious** surfaces. In 1985, Water accounted for 63.59% of the study area, but this figure declined to 39.46% by 2024, reaching a time-series low of 37.16% in 2022. Simultaneously, Barren or Impervious cover expanded from 8.73% to 30.54%. This inverse relationship represents the most significant compositional shift in the 40-year record, with the most rapid period of change occurring between 2000 and 2003, when Water cover dropped by nearly 10 percentage points.

Vegetation dynamics show a general increase in herbaceous cover and a recent decline in shrubland. **Grass/Forb/Herb** cover rose from 10.5% in 1985 to 14.08% in 2024, peaking at 15.14% in 2023. In contrast, **Shrubs** increased through the mid-2000s to a peak of 7.31% in 2007 before declining to 2.54% by 2024. Other classes remained relatively stable or minor: * **Trees** and tree-related mixes consistently represented less than 2% of the total area. * **Grass/Forb/Herb & Shrubs Mix** maintained a steady presence, fluctuating between 10% and 12% for the majority of the period.

Spatially, the transition from Water to Barren or Impervious land is concentrated along the margins of the central water body. The animated imagery shows a progressive recession of the shoreline, most notably in the northern and southern reaches of the basin. As the water level dropped, the exposed lakebed transitioned almost immediately to Barren cover. Vegetative expansion, particularly the increase in Grass/Forb/Herb, is observed primarily on the outer edges of these newly exposed areas and within the established terrestrial zones on the western and eastern peripheries.



	Trees	Grass/Forb/Herb	Barren & Grass/Forb/Herb Mix	Barren or Impervious	Water	Non-Processing Area Mask	Shrubs & Trees Mix	Grass/Forb/Herb & Trees Mix	Barren & Trees Mix	Shrubs	Grass/Forb/Herb & Shrubs Mix	Barren & Shrubs Mix	Snow or Ice
1985	0.87	10.50	0.02	8.73	63.59	0.06	0.02	0.72	0.01	3.92	11.47	0.09	0.0
1986	0.85	10.54	0.09	7.67	64.33	0.02	0.02	0.72	0.01	4.58	11.12	0.06	0.0
1987	0.88	9.45	0.09	7.88	64.21	0.01	0.02	0.67	0.01	5.23	11.46	0.07	0.0
1988	0.84	9.24	0.10	9.37	62.70	0.01	0.02	0.66	0.01	5.42	11.55	0.07	0.0
1989	0.84	8.78	0.10	12.44	59.67	0.00	0.02	0.66	0.01	5.47	11.91	0.09	0.0
1990	0.83	8.44	0.10	14.33	57.90	0.00	0.02	0.67	0.01	5.51	12.07	0.11	0.0
1991	0.80	8.47	0.11	16.56	55.59	0.00	0.02	0.62	0.01	5.44	12.26	0.11	0.0
1992	0.79	8.82	0.11	18.61	53.40	0.00	0.02	0.63	0.01	5.41	12.11	0.10	0.0
1993	0.76	10.13	0.12	18.30	53.43	0.00	0.02	0.64	0.01	5.00	11.53	0.06	0.0
1994	0.75	10.56	0.12	19.44	51.85	0.00	0.02	0.66	0.01	4.98	11.55	0.05	0.0
1995	0.75	11.18	0.13	18.93	52.10	0.00	0.02	0.66	0.01	4.23	11.95	0.05	0.0
1996	0.74	10.92	0.12	18.16	52.69	0.00	0.03	0.66	0.01	4.39	12.21	0.05	0.0
1997	0.73	10.51	0.12	17.26	53.43	0.00	0.03	0.65	0.01	4.84	12.36	0.06	0.0
1998	0.72	10.49	0.13	15.81	54.82	0.00	0.03	0.64	0.01	4.91	12.38	0.06	0.0
1999	0.71	9.92	0.12	15.55	55.06	0.00	0.03	0.57	0.01	5.37	12.61	0.06	0.0
2000	0.65	9.71	0.13	16.52	53.97	0.00	0.03	0.53	0.01	5.70	12.69	0.07	0.0
2001	0.57	9.89	0.12	19.00	51.42	0.00	0.02	0.54	0.01	5.75	12.62	0.07	0.0
2002	0.53	9.85	0.17	21.97	48.25	0.00	0.02	0.51	0.01	6.34	12.29	0.07	0.0
2003	0.53	9.85	0.17	25.72	44.47	0.00	0.02	0.54	0.01	6.53	12.08	0.07	0.0
2004	0.54	10.45	0.17	25.55	44.52	0.00	0.03	0.54	0.01	6.57	11.54	0.07	0.0
2005	0.57	11.04	0.16	24.07	45.80	0.00	0.03	0.55	0.01	6.73	10.98	0.07	0.0
2006	0.55	10.80	0.16	24.06	45.68	0.00	0.03	0.55	0.01	7.19	10.90	0.07	0.0
2007	0.52	10.79	0.17	25.52	44.05	0.00	0.03	0.55	0.01	7.31	10.98	0.08	0.0
2008	0.51	10.86	0.17	26.05	43.39	0.00	0.03	0.54	0.01	7.16	11.20	0.08	0.0
2009	0.51	11.08	0.17	25.83	43.75	0.00	0.03	0.52	0.01	7.10	10.91	0.08	0.0
2010	0.49	11.24	0.17	26.10	43.48	0.00	0.03	0.52	0.01	7.21	10.66	0.08	0.0
2011	0.51	11.74	0.18	24.64	45.04	0.00	0.03	0.53	0.01	6.68	10.55	0.09	0.0
2012	0.49	11.80	0.19	25.30	44.46	0.00	0.03	0.52	0.01	6.07	11.02	0.10	0.0
2013	0.47	12.22	0.19	26.27	43.32	0.00	0.04	0.52	0.01	5.62	11.22	0.11	0.0
2014	0.49	11.93	0.19	27.72	41.85	0.00	0.04	0.54	0.01	5.99	11.11	0.13	0.0
2015	0.51	12.52	0.19	29.11	40.27	0.00	0.04	0.55	0.01	5.87	10.82	0.13	0.0
2016	0.49	14.25	0.16	29.28	39.72	0.00	0.04	0.57	0.01	5.04	10.33	0.10	0.0
2017	0.50	14.55	0.16	28.33	40.41	0.00	0.04	0.55	0.01	4.98	10.36	0.10	0.0
2018	0.52	14.47	0.14	28.47	40.20	0.00	0.04	0.50	0.01	5.07	10.46	0.10	0.0
2019	0.54	15.07	0.12	27.67	40.85	0.00	0.05	0.55	0.01	4.72	10.32	0.09	0.0
2020	0.56	14.20	0.13	28.24	40.34	0.00	0.05	0.58	0.01	4.99	10.76	0.12	0.0
2021	0.61	13.90	0.13	31.48	37.44	0.00	0.05	0.68	0.01	4.58	10.96	0.14	0.0
2022	0.68	14.40	0.13	32.37	37.16	0.00	0.05	0.77	0.02	3.56	10.71	0.16	0.0
2023	0.88	15.14	0.10	30.84	38.94	0.00	0.07	0.80	0.02	3.06	9.97	0.20	0.0
2024	1.06	14.08	0.08	30.54	39.46	0.07	0.07	1.09	0.02	2.54	10.43	0.55	0.0

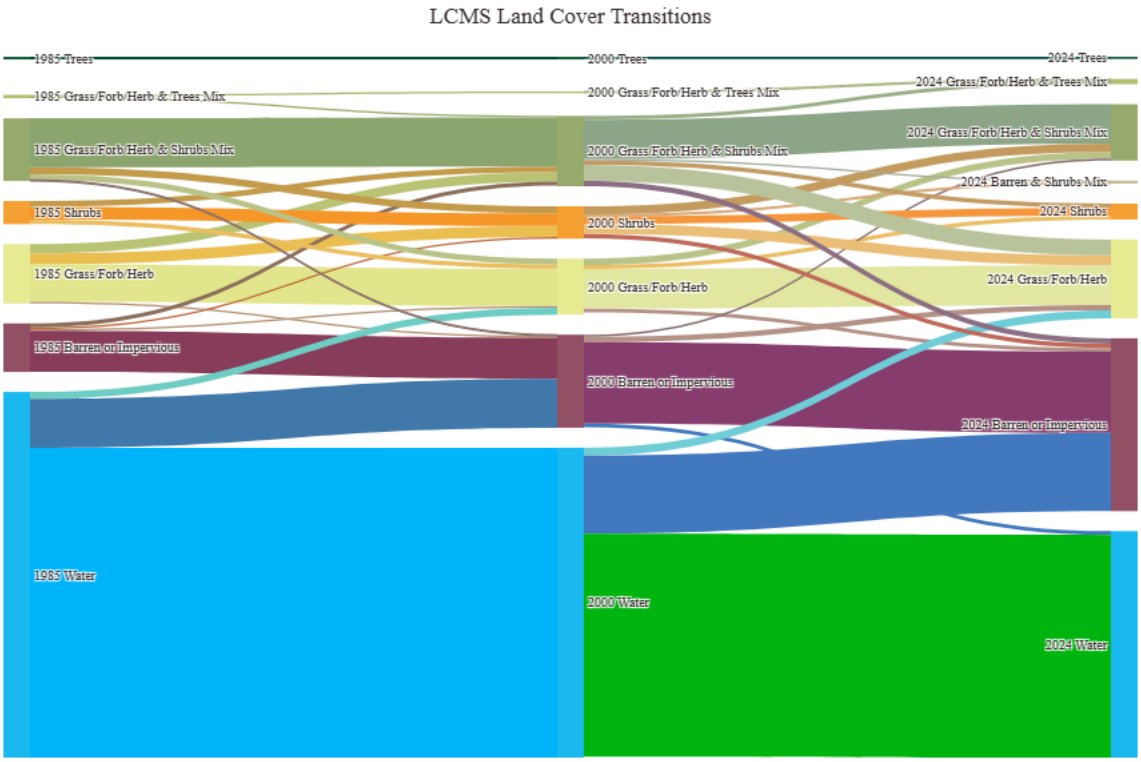
LCMS Land Cover Transitions

LCMS Land Cover Transitions

Between 1985 and 2000, the landscape exhibited the highest stability within the **Water** class, which retained 53.76% of the total area. The most frequent transition between different classes was the conversion of **Water to Barren or Impervious**, accounting for 8.47% of the area. Other notable stable classes during this period included **Grass/Forb/Herb & Shrubs Mix** (8.49%) and **Barren or Impervious** (7.14%). Vegetation dynamics were characterized by a shift from **Grass/Forb/Herb** toward shrub-dominated classes, with 1.88% transitioning to **Shrubs** and 1.61% to **Grass/Forb/Herb & Shrubs Mix**.

From 2000 to 2024, the stability of the **Water** class decreased to 38.65%, while the persistence of **Barren or Impervious** surfaces increased to 14.1%. The transition from **Water to Barren or Impervious** remained a primary driver of change, involving 13.56% of the total area. Within vegetated classes, there was a measurable shift toward herbaceous cover: * **Grass/Forb/Herb & Shrubs Mix** to **Grass/Forb/Herb**: 2.77% * **Shrubs** to **Grass/Forb/Herb**: 1.72% * **Shrubs** to **Grass/Forb/Herb & Shrubs Mix**: 1.49%

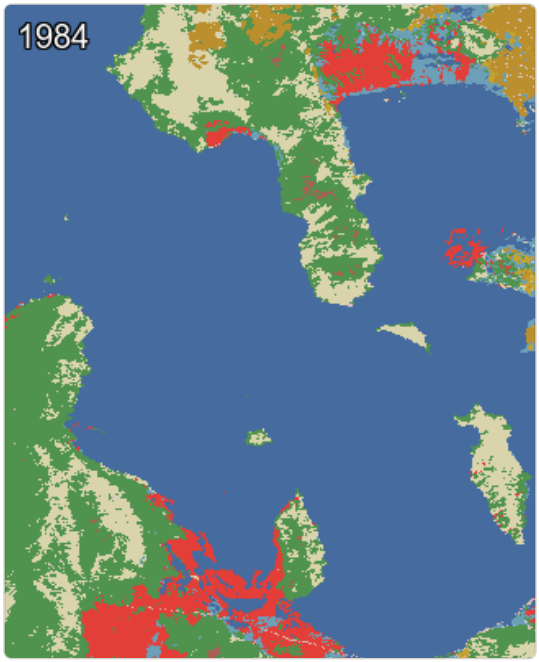
The long-term data indicates a consistent trend of water loss and the expansion of barren or impervious areas. While classes such as **Trees** and **Shrubs & Trees Mix** remained stable but occupied a small percentage of the total area (less than 0.5% each), the **Grass/Forb/Herb** and **Shrubs** complexes showed the highest rates of internal flux. By 2024, the **Barren or Impervious** class represented a significantly larger portion of the landscape than in the 1985–2000 interval, driven by both class persistence and transitions from the **Water** and **Grass/Forb/Herb** categories.



	Trees 2000	Tall Shrubs & Trees Mix (AK Only) 2000	Shrubs & Trees Mix 2000	Grass/Forb/Herb & Trees Mix 2000	Barren & Trees Mix 2000	Tall Shrubs (AK Only) 2000	Shrubs 2000	Grass/Forb/Herb & Shrubs Mix 2000	Barren & Shrubs Mix 2000	Grass/Forb/Herb 2000	Barren & Grass/Forb/Herb Mix 2000	Barren or Impervious 2000	Snow or Ice 2000	Water 2000	Non-Processing Area Mask 2000	Trees 2024	Tall Shrubs & Trees Mix (AK Only) 2024	Shrubs & Trees Mix 2024
Trees 1985	0.47	0.0	0.01	0.10	0.00	0.0	0.07	0.16	0.00	0.03	0.00	0.04	0.0	0.00	0.0	NaN	NaN	NaN
Tall Shrubs & Trees Mix (AK Only) 1985	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.0	NaN	NaN	NaN
Shrubs & Trees Mix 1985	0.00	0.0	0.01	0.00	0.00	0.0	0.01	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.0	NaN	NaN	NaN
Grass/Forb/Herb & Trees Mix 1985	0.06	0.0	0.01	0.31	0.00	0.0	0.10	0.28	0.00	0.01	0.00	0.03	0.0	0.00	0.0	NaN	NaN	NaN
Barren & Trees Mix 1985	0.00	0.0	0.00	0.00	0.01	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.0	NaN	NaN	NaN
Tall Shrubs (AK Only) 1985	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.0	NaN	NaN	NaN
Shrubs 1985	0.01	0.0	0.01	0.04	0.00	0.0	2.21	1.01	0.02	0.79	0.00	0.18	0.0	0.00	0.0	NaN	NaN	NaN
Grass/Forb/Herb & Shrubs Mix 1985	0.02	0.0	0.00	0.09	0.00	0.0	1.21	8.49	0.05	0.97	0.00	0.21	0.0	0.00	0.0	NaN	NaN	NaN

	Trees 2000	Tall Shrubs & Trees Mix (AK Only) 2000	Shrubs & Trees Mix 2000	Grass/Forb/Herb & Trees Mix 2000	Barren & Trees Mix 2000	Tall Shrubs (AK Only) 2000	Shrubs 2000	Grass/Forb/Herb & Shrubs Mix 2000	Barren & Shrubs Mix 2000	Grass/Forb/Herb 2000	Barren & Grass/Forb/Herb Mix 2000	Barren or Impervious 2000	Snow or Ice 2000	Water 2000	Non-Processing Area Mask 2000	Trees 2024	Tall Shrubs & Trees Mix (AK Only) 2024	Shrubs & Trees Mix 2024
Barren & Shrubs Mix 1985	0.00	0.0	0.00	0.00	0.00	0.0	0.04	0.07	0.03	0.02	0.00	0.02	0.0	0.00	0.0	NaN	NaN	NaN
Grass/Forb/Herb 1985	0.03	0.0	0.00	0.01	0.00	0.0	1.88	1.61	0.02	6.48	0.00	0.32	0.0	0.02	0.0	NaN	NaN	NaN
Barren & Grass/Forb/Herb Mix 1985	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.03	0.01	0.0	0.00	0.0	NaN	NaN	NaN
Barren or Impervious 1985	0.02	0.0	0.00	0.03	0.00	0.0	0.31	0.69	0.03	0.27	0.13	7.14	0.0	0.12	0.0	NaN	NaN	NaN
Snow or Ice 1985	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.0	NaN	NaN	NaN
Water 1985	0.06	0.0	0.00	0.00	0.00	0.0	0.02	0.01	0.00	1.21	0.00	8.47	0.0	53.76	0.0	NaN	NaN	NaN
Non-Processing Area Mask 1985	0.00	0.0	0.00	0.00	0.00	0.0	0.00	0.01	0.00	0.02	0.00	0.05	0.0	0.00	0.0	NaN	NaN	NaN
Trees 2000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	0.38	0.0	0.01
Tall Shrubs & Trees Mix (AK Only) 2000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	0.00	0.0	0.00
Shrubs & Trees Mix 2000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	0.01	0.0	0.01
Grass/Forb/Herb & Trees Mix 2000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	0.14	0.0	0.02
Barren & Trees Mix 2000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	0.00	0.0	0.00
Tall Shrubs (AK Only) 2000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	0.00	0.0	0.00
Shrubs 2000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	0.08	0.0	0.03
Grass/Forb/Herb & Shrubs Mix 2000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	0.08	0.0	0.03
Barren & Shrubs Mix 2000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	0.00	0.0	0.00
Grass/Forb/Herb 2000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	0.19	0.0	0.00
Barren & Grass/Forb/Herb Mix 2000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	0.00	0.0	0.00
Barren or Impervious 2000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	0.09	0.0	0.00
Snow or Ice 2000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	0.00	0.0	0.00
Water 2000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	0.17	0.0	0.00
Non-Processing Area Mask 2000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	0.00	0.0	0.00

NLCD Land Cover (1985-2024)

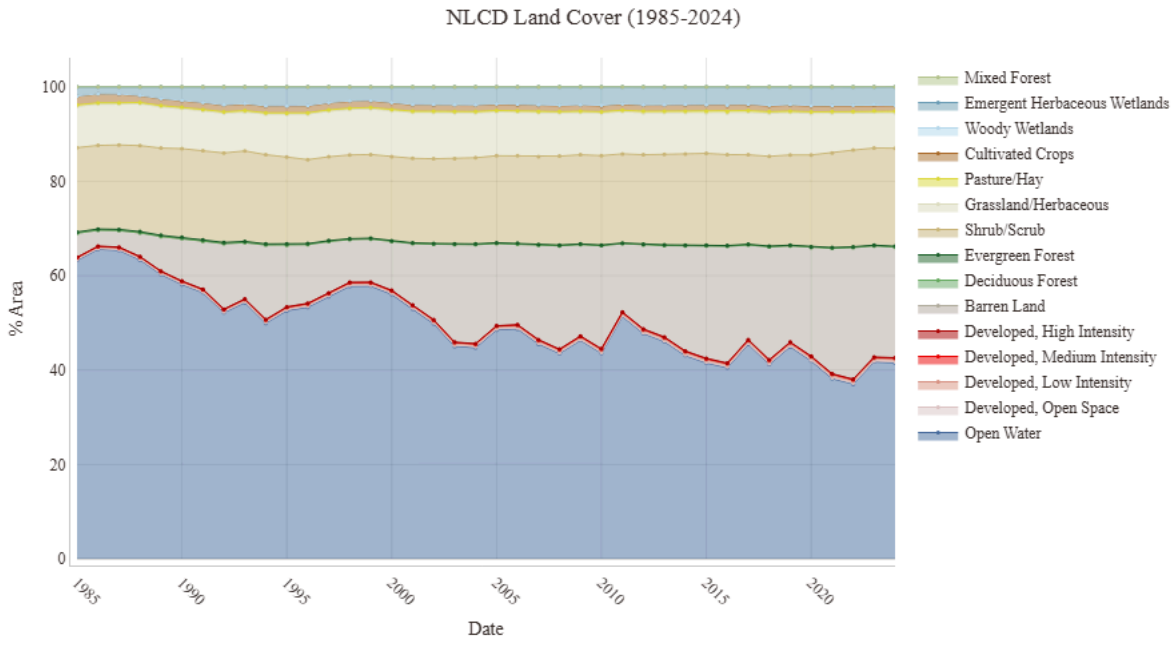


NLCD Land Cover (1985-2024)

The study area is characterized by a dominant and highly dynamic relationship between **Open Water** and **Barren Land**. Between 1985 and 2024, Open Water decreased from a peak of 65.59% (1986) to 41.55% (2024), reaching a historical low of 37.04% in 2022. This decline is directly offset by the expansion of Barren Land, which grew from 5.11% in 1985 to 23.47% in 2024. The spatial data confirms that this transition occurs primarily along the receding margins of the central water body, indicating a significant and persistent contraction of the shoreline over the 40-year period.

Vegetative cover and anthropogenic development show more gradual but steady trends: * **Shrub/Scrub** remains the primary terrestrial land cover, increasing from 17.87% to 20.78%. * **Grassland/Herbaceous** and **Cultivated Crops** both experienced net declines, ending the period at 7.47% and 1.20%, respectively. * **Developed** classes (Open Space through High Intensity) represent a small portion of the total area but showed consistent growth, nearly doubling from an aggregate of 0.64% in 1985 to 1.05% in 2024. * **Emergent Herbaceous Wetlands** exhibited high inter-annual variability, fluctuating between 1.46% and 4.05% in response to changing water levels.

Spatially, the most significant changes are concentrated in the basin interior, where the central water body has fragmented and retreated, leaving behind expansive rings of Barren Land. Developed areas are concentrated in the northeastern and southwestern peripheries, showing incremental outward expansion into adjacent Grassland and Shrub/Scrub zones. The Shrub/Scrub class maintains a stable presence in the higher-elevation upland areas, while the wetland and barren classes remain highly sensitive to the fluctuating perimeter of the Open Water.



	Open Water	Developed, Open Space	Developed, Low Intensity	Developed, Medium Intensity	Developed, High Intensity	Barren Land	Deciduous Forest	Evergreen Forest	Shrub/Scrub	Grassland/Herbaceous	Pasture/Hay	Cultivated Crops	Woody Wetlands	Emergent Herbaceous Wetlands	Mixed Forest
1985	63.25	0.14	0.30	0.15	0.05	5.11	0.0	0.25	17.87	8.73	0.32	1.83	0.05	1.94	0.0
1986	65.59	0.15	0.31	0.15	0.05	3.39	0.0	0.25	17.72	8.71	0.30	1.87	0.05	1.46	0.0
1987	65.38	0.16	0.31	0.15	0.05	3.51	0.0	0.25	17.89	8.63	0.29	1.82	0.05	1.49	0.0
1988	63.36	0.16	0.32	0.16	0.05	5.03	0.0	0.25	18.23	8.84	0.29	1.44	0.06	1.82	0.0
1989	60.28	0.16	0.32	0.16	0.05	7.33	0.0	0.25	18.48	8.72	0.29	1.42	0.07	2.47	0.0
1990	58.14	0.17	0.33	0.17	0.05	9.03	0.0	0.25	18.78	8.48	0.29	1.43	0.07	2.81	0.0
1991	56.40	0.17	0.33	0.18	0.05	10.20	0.0	0.25	18.88	8.47	0.29	1.41	0.07	3.30	0.0
1992	52.16	0.17	0.34	0.18	0.05	13.88	0.0	0.25	18.97	8.40	0.30	1.39	0.08	3.83	0.0
1993	54.31	0.18	0.34	0.18	0.05	11.95	0.0	0.25	19.17	8.23	0.31	1.41	0.08	3.54	0.0
1994	49.93	0.18	0.35	0.19	0.06	15.79	0.0	0.23	18.90	8.57	0.30	1.41	0.08	4.01	0.0
1995	52.59	0.19	0.35	0.19	0.06	13.11	0.0	0.22	18.42	9.00	0.30	1.43	0.08	4.05	0.0
1996	53.32	0.20	0.35	0.20	0.06	12.46	0.0	0.20	17.80	9.62	0.30	1.45	0.08	3.97	0.0
1997	55.55	0.20	0.35	0.20	0.06	10.88	0.0	0.18	17.77	9.64	0.30	1.45	0.08	3.34	0.0
1998	57.77	0.20	0.35	0.21	0.06	9.07	0.0	0.18	17.74	9.62	0.30	1.46	0.08	2.96	0.0
1999	57.80	0.20	0.34	0.22	0.06	9.14	0.0	0.18	17.73	9.65	0.29	1.44	0.08	2.86	0.0
2000	56.06	0.21	0.34	0.22	0.06	10.34	0.0	0.18	17.82	9.68	0.29	1.39	0.08	3.33	0.0
2001	52.92	0.21	0.35	0.22	0.06	13.03	0.0	0.16	17.91	9.69	0.29	1.36	0.08	3.71	0.0
2002	49.83	0.20	0.35	0.23	0.06	16.01	0.0	0.15	17.94	9.77	0.29	1.32	0.09	3.75	0.0
2003	45.05	0.20	0.36	0.23	0.06	20.70	0.0	0.15	18.05	9.71	0.29	1.32	0.09	3.79	0.0
2004	44.74	0.20	0.37	0.23	0.06	20.98	0.0	0.15	18.26	9.48	0.30	1.32	0.09	3.80	0.0
2005	48.51	0.19	0.38	0.24	0.06	17.43	0.0	0.16	18.44	9.28	0.30	1.32	0.09	3.59	0.0
2006	48.73	0.19	0.38	0.25	0.06	17.09	0.0	0.16	18.55	9.19	0.30	1.31	0.09	3.70	0.0
2007	45.54	0.18	0.39	0.26	0.07	20.00	0.0	0.15	18.66	9.24	0.29	1.30	0.09	3.84	0.0
2008	43.52	0.17	0.40	0.26	0.07	21.92	0.0	0.15	18.84	9.13	0.29	1.27	0.09	3.89	0.0
2009	46.30	0.16	0.41	0.26	0.08	19.37	0.0	0.14	18.90	8.94	0.29	1.28	0.09	3.77	0.0
2010	43.57	0.16	0.42	0.27	0.08	21.86	0.0	0.13	18.95	8.99	0.30	1.27	0.09	3.91	0.0
2011	51.36	0.16	0.42	0.27	0.08	14.49	0.0	0.14	18.89	8.93	0.31	1.27	0.10	3.58	0.0
2012	47.76	0.16	0.43	0.27	0.08	17.81	0.0	0.14	18.98	8.89	0.31	1.26	0.10	3.79	0.0
2013	46.05	0.17	0.43	0.28	0.08	19.38	0.0	0.14	19.18	8.82	0.32	1.26	0.11	3.79	0.0
2014	43.07	0.17	0.44	0.27	0.08	22.32	0.0	0.14	19.27	8.78	0.32	1.26	0.11	3.77	0.0
2015	41.52	0.17	0.44	0.27	0.08	23.85	0.0	0.14	19.44	8.61	0.33	1.26	0.11	3.78	0.0
2016	40.50	0.18	0.44	0.27	0.08	24.76	0.0	0.14	19.29	8.74	0.34	1.26	0.11	3.89	0.0
2017	45.47	0.18	0.44	0.28	0.08	20.07	0.0	0.14	18.95	8.99	0.36	1.26	0.11	3.68	0.0
2018	41.19	0.18	0.44	0.28	0.08	23.95	0.0	0.14	19.03	9.10	0.35	1.24	0.11	3.90	0.0
2019	44.95	0.18	0.44	0.28	0.08	20.39	0.0	0.14	19.12	8.95	0.34	1.26	0.11	3.76	0.0
2020	41.95	0.19	0.44	0.28	0.08	23.03	0.0	0.14	19.47	8.78	0.34	1.25	0.11	3.94	0.0
2021	38.20	0.19	0.45	0.28	0.09	26.60	0.0	0.14	20.04	8.44	0.31	1.22	0.12	3.92	0.0
2022	37.04	0.19	0.46	0.28	0.09	27.91	0.0	0.14	20.49	7.93	0.27	1.22	0.12	3.86	0.0
2023	41.73	0.19	0.47	0.28	0.09	23.56	0.0	0.14	20.58	7.57	0.32	1.24	0.13	3.69	0.0
2024	41.55	0.20	0.47	0.29	0.09	23.47	0.0	0.15	20.78	7.47	0.35	1.20	0.12	3.87	0.0

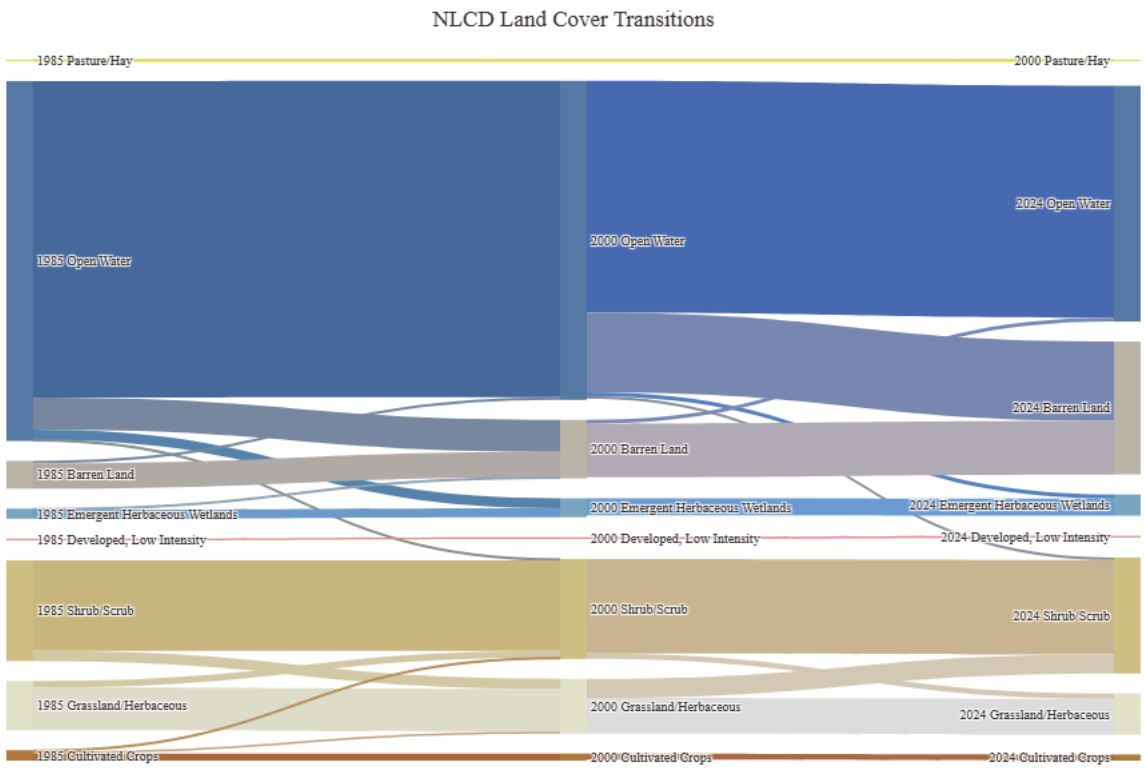
NLCD Land Cover Transitions

NLCD Land Cover Transitions

Analysis of land cover transitions between 1985, 2000, and 2024 reveals a landscape dominated by **Open Water**, **Shrub/Scrub**, and **Barren Land**, with significant flux occurring between these categories. From 1985 to 2000, the most substantial stable class was Open Water, with 55.54% of the total area remaining unchanged. During this same period, 15.89% of the landscape persisted as Shrub/Scrub. The most notable transition involved the conversion of 5.63% of 1985 Open Water into Barren Land by 2000.

The transition period from 2000 to 2024 indicates an acceleration in the conversion of aquatic features to terrestrial classes. The area of persistent **Open Water** decreased to 40.7%, while the transition from Open Water to **Barren Land** increased significantly to 13.94%. Simultaneously, **Shrub/Scrub** showed high retention at 16.57%, supplemented by a 3.36% transition from Grassland/Herbaceous areas. **Emergent Herbaceous Wetlands** also saw a notable shift, with 3.0% of the 2000 extent persisting through 2024, while 0.73% of former Open Water transitioned into this category.

Developed and agricultural classes represent a small but stable fraction of the total land cover. **Developed, Low Intensity** areas maintained a consistent footprint, with 0.29% persistence between 1985–2000 and 0.31% between 2000–2024. **Cultivated Crops** and **Pasture/Hay** exhibited minimal change, with the majority of these classes remaining stable or undergoing minor internal exchanges with Grassland/Herbaceous and Shrub/Scrub categories. Overall, the data reflects a trend of receding water levels and the subsequent expansion of barren and shrub-dominated landscapes.



	Open Water 2000	Perennial Ice/Snow 2000	Developed, Open Space 2000	Developed, Low Intensity 2000	Developed, Medium Intensity 2000	Developed, High Intensity 2000	Barren Land 2000	Deciduous Forest 2000	Evergreen Forest 2000	Mixed Forest 2000	Shrub/Scrub 2000	Grassland/Herbaceous 2000	Pasture/Hay 2000	Cultivated Crops 2000	Woody Wetlands 2000	Emerg Herbac Wetla 200
Open Water 1985	55.54	0.0	0.00	0.02	0.04	0.00	5.63	0.0	0.00	0.0	0.23	0.01	0.01	0.00	0.02	
Perennial Ice/Snow 1985	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.00	0.00	
Developed, Open Space 1985	0.00	0.0	0.14	0.01	0.00	0.00	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.00	0.00	
Developed, Low Intensity 1985	0.00	0.0	0.00	0.29	0.01	0.00	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.00	0.00	
Developed, Medium Intensity 1985	0.00	0.0	0.00	0.00	0.15	0.00	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.00	0.00	
Developed, High Intensity 1985	0.00	0.0	0.00	0.00	0.00	0.05	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.00	0.00	
Barren Land 1985	0.45	0.0	0.00	0.00	0.00	0.00	4.42	0.0	0.00	0.0	0.16	0.04	0.00	0.00	0.00	
Deciduous Forest 1985	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.00	0.00	
Evergreen Forest 1985	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.0	0.16	0.0	0.06	0.03	0.00	0.00	0.00	
Mixed Forest 1985	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.00	0.00	
Shrub/Scrub 1985	0.01	0.0	0.03	0.02	0.01	0.01	0.06	0.0	0.02	0.0	15.89	1.77	0.01	0.03	0.00	
Grassland/Herbaceous 1985	0.00	0.0	0.03	0.00	0.00	0.00	0.01	0.0	0.00	0.0	1.16	7.47	0.01	0.06	0.00	

	Open Water 2000	Perennial Ice/Snow 2000	Developed, Open Space 2000	Developed, Low Intensity 2000	Developed, Medium Intensity 2000	Developed, High Intensity 2000	Barren Land 2000	Deciduous Forest 2000	Evergreen Forest 2000	Mixed Forest 2000	Shrub/Scrub 2000	Grassland/Herbaceous 2000	Pasture/Hay 2000	Cultivated Crops 2000	Woody Wetlands 2000	Emerg Herbac Wetla 200
Pasture/Hay 1985	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.0	0.01	0.02	0.23	0.05	0.00	
Cultivated Crops 1985	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.0	0.23	0.35	0.02	1.24	0.00	
Woody Wetlands 1985	0.00	0.0	0.00	0.00	0.00	0.00	0.00	0.0	0.00	0.0	0.00	0.00	0.00	0.00	0.04	
Emergent Herbaceous Wetlands 1985	0.05	0.0	0.00	0.00	0.00	0.00	0.21	0.0	0.00	0.0	0.06	0.00	0.02	0.02	0.02	
Open Water 2000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
Perennial Ice/Snow 2000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
Developed, Open Space 2000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
Developed, Low Intensity 2000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
Developed, Medium Intensity 2000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
Developed, High Intensity 2000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
Barren Land 2000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
Deciduous Forest 2000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
Evergreen Forest 2000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
Mixed Forest 2000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
Shrub/Scrub 2000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
Grassland/Herbaceous 2000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
Pasture/Hay 2000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
Cultivated Crops 2000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
Woody Wetlands 2000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
Emergent Herbaceous Wetlands 2000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	